

Dhruv Athaide

Android Engineer · Kotlin, Java & Flutter · Firebase & Mobile Architecture · Published Researcher

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PROFESSIONAL SUMMARY

Android Engineer with **2+ years** shipping production Android and Flutter apps, including **3 apps independently taken from concept to Google Play**. Owns architecture, sprint planning, and release pipelines end-to-end; has mentored **25+ developers** and led cross-functional teams of **60+**. **Published researcher** - co-author of a peer-reviewed IGI Global book chapter (**©2026**) on Android vulnerability assessment and mobile threat monitoring.

TECHNICAL SKILLS

Android Development: Kotlin, Java, Jetpack Compose, Flutter, Dart, Android SDK, Firebase, MVVM, Clean Architecture, Room, WorkManager, Coroutines & Flow, Retrofit, OkHttp, Hilt, Play Billing, Material Design, Gradle, REST APIs

Engineering Practices: OOP & Design Patterns, Data Structures & Algorithms, Multithreading, Memory Management, Unit & Integration Testing, Code Review, Agile Sprint Planning, Low-Level Design (LLD)

Security & Systems: Encrypted Storage, Biometric Auth, OWASP Mobile Top 10, Applied Cryptography, Android Internals (AOSP, SELinux, JNI/NDK)

Cloud & Tools: Firebase (FCM, Firestore, Auth), Node.js, Azure, Docker, Git, CI/CD, MongoDB

WORK EXPERIENCE

DeepCytes Cyber Labs Mumbai, Maharashtra

August 2025 - Present

Android Security Engineer

- Re-architected a production scanning engine, cutting scan latency **30%** and main-thread blocking **85%** with zero cloud dependency.
- Owned architecture and delivery for a modular Android security toolkit adopted across **10+ enterprise engagements**, improving assessment fidelity **40%**.
- Hardened a custom Android distribution (AOSP-based), reducing OS attack surface **65%** for high-assurance deployments.
- Reduced false positives **35%** across **10,000+ device samples** in a BLE-based tracker-detection system through spatial-temporal movement analysis.

DeepCytes Cyber Labs Mumbai, Maharashtra

January 2024 - June 2025

Mobile Development Team Lead Intern

- Owned the Mobile Development Vertical (Jan-Jun 2025): drove strategy, sprint planning, code reviews, and Play Store release pipelines for **5 Android products**; mentored **25+ developers** to production in 3 months, cutting PR rework **20%** with **zero critical post-release defects**.
- Directed a cross-functional team of **60+ engineers and researchers**, lifting project completion rates **15%** and delivery timelines **25%** through sprint-velocity tracking.
- Delivered **4 enterprise platforms**, including a compliance LMS serving **500+ active users** and a multi-tenant SaaS product, lifting client engagement **35%** and customer acquisition **45%**.

Truboard Partners Mumbai, Maharashtra

June 2024 - August 2024

Production & Software Testing Intern

- Cut erroneous forecasts an estimated **15%** by fine-tuning a B2B forecasting model.
- Cut fleet management costs an estimated **20%** by building a real-time asset-monitoring system tracking **200+ vehicles**.

DeepCytes Cyber Labs Mumbai, Maharashtra

June 2023 - December 2023

AI & Deep Web Intern

- Improved the accuracy of cyber-intelligence and research output **10%** by engineering AI-based intelligence-gathering tools and research pipelines.

KEY PROJECTS

Guardient - Mobile Privacy & Threat Intelligence App (Google Play · DC Webstore)

Kotlin, Android SDK, C++ (NDK), MVVM

- Architected a modular Clean Architecture/MVVM codebase with a heuristic scoring engine evaluating permission synergies and installer trust.
- Built a custom-animated risk dashboard (donut and HUD-style charts) with state-hashing to avoid redundant re-renders, keeping the UI fluid while scanning **100+ installed apps** in real time.
- Detected **100%** of malware and spyware samples across a 10–20 app test suite, entirely on-device, with **30%** lower scan latency and shipped to Google Play

LockGuard+ - Intruder Detection & Secure Evidence Management App (Google Play · DC Webstore)

Kotlin, CameraX, C++ (NDK/JNI), AES-256

Security app that silently captures intruder photos on failed unlock attempts and delivers encrypted evidence alerts in real time, monetised via subscription.

- Built a headless CameraX capture pipeline with a custom lifecycle handler, achieving **100%** alert reliability with **sub-500ms** delivery latency.
- Independently shipped a self-contained subscription product to Google Play — architecture, billing integration, and release, solo, end-to-end.

NoSurveil - BLE Anti-Stalking & Tracker Detection App (Google Play · DC Webstore)

Kotlin, Android SDK, Room, Hilt, C++ (NDK), MVVM

Privacy-first app that detects physical stalking by correlating a user's movement against nearby Bluetooth device patterns over time.

- Designed an $O(N+M)$ temporal-alignment algorithm correlating location histories, reducing false positives **35%** across **10,000+ device samples**.
- Built a battery-optimized background BLE scanning engine, keeping continuous operation below **2% battery/hour**; shipped with zero-cloud, privacy-by-design architecture.

BlackoutMode - Enterprise Mobile Privacy Hardening Platform (DC Webstore)

Kotlin, Jetpack Compose, C++ (NDK/JNI), Bouncy Castle (Post-Quantum ML-KEM), AES-256

Enterprise Android platform providing active hardware-resource protection (camera/microphone lockout), on-device DNS-based traffic filtering, and a post-quantum encrypted vault for sensitive enterprise data.

- Built a thread-safe hardware-protection service blocking **90%** of unauthorized background camera and microphone access attempts.
- Implemented a quantum-safe encrypted vault (Bouncy Castle ML-KEM key encapsulation, HKDF-derived AES-256) with chunked streaming encryption for large-file support.

Torch - Enterprise Remote Data Protection & Device Management Platform (Website)

Kotlin, Jetpack Compose, C++ (NDK/JNI), Node.js/Express, React, Firebase (FCM), MongoDB

Full-stack enterprise mobile device management (MDM) system enabling authorized administrators to remotely track, lock, and irrecoverably wipe compromised or lost devices, backed by an Android client, a Node.js/Express API, and a React admin dashboard.

- Engineered a cryptographic-erasure engine that rotating across 5 symmetric ciphers to shred files up to 25GB in-RAM before disk overwrite, ensuring forensic unrecoverability while minimizing flash-storage wear.
- Built a scalable fleet heartbeat service tracking online/offline status across a 100-device fleet, with automatic offline-state pruning.

Flash - Enterprise Mobile Device Management (MDM) Suite (Website)

Kotlin, Jetpack Compose, Node.js/Express, MongoDB, Firebase

Enterprise MDM platform enabling organizations to remotely locate, encrypt, or wipe corporate Android devices, with hierarchical role-based access control for organization and branch administrators.

- Implemented in-place AES-256-GCM file encryption via the Storage Access Framework, with dynamic disk-space checks preventing OS-level failures on encryption jobs up to 20GB.
- Designed a hierarchical RBAC system letting org/branch admins securely act on devices within their assigned scope, verified dynamically on every request.

DeepCloak - Mobile Security & Device Diagnostics App (Google Play · DC Webstore)

Flutter, Dart, C++ (JNI), Hive, flutter_secure_storage

Cross-platform privacy and device-diagnostics app providing real-time hardware/battery telemetry, a network diagnostics toolkit, and a cryptographically enforced enterprise licensing gate.

- Built a fail-closed licensing framework by injecting JWT validation at the app-root level, eliminating bypass risk from deep links or routing race conditions.
- Engineered an on-device telemetry engine (Hive local storage) generating predictive device-health alerts without cloud analytics, preserving user privacy.

Cyber Investigator Battlegrounds (CIB) - Investigative Puzzle & Simulation Game (Google Play · DC Webstore)

Flutter, Dart, Firebase, Dio

Investigative simulation game where players analyze evidence and solve technical mini-games, backed by a reactive state machine and subscription-gated licensing.

- Built a reactive case-engine state machine with precomputed O(1) lookup maps, eliminating UI lag during rapid gameplay state changes.
- Engineered a cache-first authentication layer using API interceptors with single-flight token refresh, preventing race conditions across concurrent requests.

MobileSentinel - On-Device Mobile VAPT & Network Analysis Tool (Independent Project)

Java, Android SDK, Python, On-Device ML

Self-initiated Android penetration-testing utility built independently prior to professional engagements; its design and findings directly underpin the peer-reviewed IGI Global book chapter below.

- Delivered a complete on-device VAPT workflow with no external tooling and zero cloud dependency.
- On-device ML phishing detection achieved **94%+ accuracy** with **sub-50ms** inference latency.

RESEARCH & PUBLICATIONS

Design of Comprehensive Mobile Device Vulnerability Checking Application

March 2026

Book Chapter, Ch. 6 · Advanced Cybersecurity for Threats Exploitation and Digital Risk · IGI Global Scientific Publishing

M. H. Devare[†], **D. J. Athaide**, A. M. Devare & V. Nagpal | Amity University Mumbai. Presents a comprehensive Android VAPT application design covering on-device vulnerability assessment, Wi-Fi security auditing, and real-time threat monitoring.

DOI: 10.4018/979-8-2600-0747-1.ch006 **ORCID:** orcid.org/0009-0000-3554-9798

CERTIFICATIONS

- Google Cloud Computing Foundations Certificate - Credly Verified
- Google Developer Student Club Web Developer Certificate